



COMSATS UNIVERSITY

ISLAMABAD

DESCRIPTION OF PROJECT

Final Semester Project

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HOTEL MANAGEMENT SYSTEM

Description:

This hotel management system includes features used for managing: room management, guest managing, booking, managing staff, housekeeping, and billing. The EER design includes features: specialization, weak entities, and advanced attribute types. These make sure that the database looks like your / may be used in the real world.

1. Core Entities and Attributes

HOTEL

Primary Key: The hotel entity is primarily identified by the HotelID.

Attribute: The hotel entity can also have standard hotel entity characteristics including StarRating, HotelName, and Phone. Due to a hotel having more than one phone number, the Phone attribute is multivalued (i.e., the hotel can have multiple phone numbers). The Address attribute is composite (i.e. address is comprised of a City and Country). Finally, the Occupancy attribute is a derived attribute that indicates how many rooms were used to determine occupancy. It is not stored directly as an attribute but defined by the rooms used at the time of occupancy.

ROOM

Primary Key: The room entity is identified by the RoomID.

Attributes: The room entity can also include RoomType, PricePerNight, Status and RoomNumber, as well. The Status attribute denotes the current condition of the room, e.g., the status of the room can either be Available or Occupied. Similar to the hotel entity's address, the room entity's RoomLocation is composite (i.e. RoomLocation is composed of the building and the RoomWing).

The Room entity is a strong entity in the constant 1:N relationship with the Hotel entity (one hotel can have many rooms). Also, similar to the hotel entity of disjoint-specialization (see section Disjoint Specialization), the room entity is further divided into subtypes:

SingleRoom (includes BedType and MaxGuests)

DoubleRoom (includes BedCount)

Suite (includes LoungeArea and Jacuzzi).

AMENITY

The AMENITY entity uses AmenityID as its primary key and has several attributes including the amenity name and category. The relationship between the ROOM and AMENITY entities is a many-to-many (M:N) relationship — meaning that one room can have more than one amenity assigned to it and more than one room can use one amenity in common. The M:N relationship is created using an associative entity.

RoomStatusLog (Weak Entity)

The Room Status Log is classified as a weak entity because it derives its existence from ROOM. For example, it cannot have a complete primary key and needs Log ID and Room ID for identification purposes. The relationship between ROOM and ROOM STATUS LOG is classified as an Identifying relationship (double diamond); where ROOM STATUS LOG has full participation. The Room Status Log also allows for the tracking of change over time, since it stores historical records such as Status and Changed At; where as, ROOM.Status provides a representation of the present status of the ROOM, Room Status Log enables tracking of the change over time, which is important for purposes of audit and performance analysis.

GUEST

The GUEST entity is defined as a strong entity by its unique identifier (Guest ID); and consists of a composite attribute (Guest Name) and a multi-valued attribute (Phone Number). Other attributes for the GUEST would include Email Address, Nationality, Age and Date of Birth.

The GUEST entity can also be divided into specializations:

- Walk In Guest (Arrival Date)
- Member Guest (Membership ID, Membership Type, Points Earned)

This allows differentiation between the casual and the Registered Customer.

Both examples of disjoint specialization illustrate how room entities relate to a single type of room in the real world.

BOOKING

The first strong central entity in the database is BOOKING which contains the Booking ID value. It's primary key. The rest of the attributes are Check In Date, Check Out Date, Booking Date, Status, Total Amount and Stay Duration. Relationships between entities are as follows; GUEST — MAKES — BOOKING (Total participation of 1 to N). BOOKING — ASSIGNS — ROOM (One to many (or many to one) with respect to the number of rooms associated with the booking), therefore, reflects real world booking scenarios.

INVOICE

The INVOICE entity is a strong entity. There are several attributes for this entity: IssueDate, AmountDue, PaymentMode, and paidFlag.

- (1:1 between INVOICE and BOOKING (a booking generates an invoice))

STAFF and Specialization

STAFF: The STAFF entity has several attributes that define how a staff member will be employed by the hotel; for example: staffID (primary key), name (composite) of the staff member, email address of the staff member, phone number (multivalued), hireDate of the staff member, salary for the staff member, and MULTI-VALUE; this last parameter is needed to effectively complete a staff member's employment duties.

The staff entity provides several classifications of job roles, including:

- Housekeeping Staff – Shift, and Experience
- Front Desk Staff – Desk#, ShiftTime
- Maintenance Staff – Specialty, Certification

This classification models real-world job roles at a hotel.

StaffProfile

The StaffProfile is a strong entity with Profile ID as its primary key and contains the Username, Password Hash, and Access Level fields. The StaffProfile has a One to One relationship with STAFF and has total participation on the StaffProfile side, and partial participation on STAFF side; meaning that every StaffProfile must belong to a STAFF member but not all STAFF members will have a StaffProfile. In addition to having a one-to-one relationship with the staff profile, a staff member employs a staff member.

HousekeepingTask

The HousekeepingTask is a strong entity that gets identified by the Task ID. The HousekeepingTask has attributes of Priority, Scheduled At, Completed At, Status, and Notes fields.

Relationships:

- Housekeeping Staff -PERFORMS Housekeeping Task (M:N, total participation on task).
- ROOM - HAS Task - Housekeeping Task (M:N, partial participation).

Multiple staff members may handle tasks, and multiple rooms will either require cleaning or not.

Relationships Summary

- HOTEL - ROOM 1:N
- ROOM - AMENITY M:N
- ROOM - RoomStatusLog 1:N (indicating, weak)
- GUEST - BOOKING 1:N
- BOOKING - ROOM M:N
- BOOKING - INVOICE 1:1
- Individual STAFF member sub-types handle specific operations:
 - o Housekeeping Staff - Task - M:N
 - o Front Desk Staff - Booking - 1:N
 - o Maintenance Staff - Room - M:N

Attribute Types Used

- Simple attributes: Hotel Name, Salary
- Composite attributes: Name, Address, Room Location
- Multi-valued attributes: Phone, Language
- Derived attributes: Occupancy • Key attributes: Hotel ID, Room ID, Etc.
- Partial Key: Log ID (for weak entity).

2. Real-World Justification

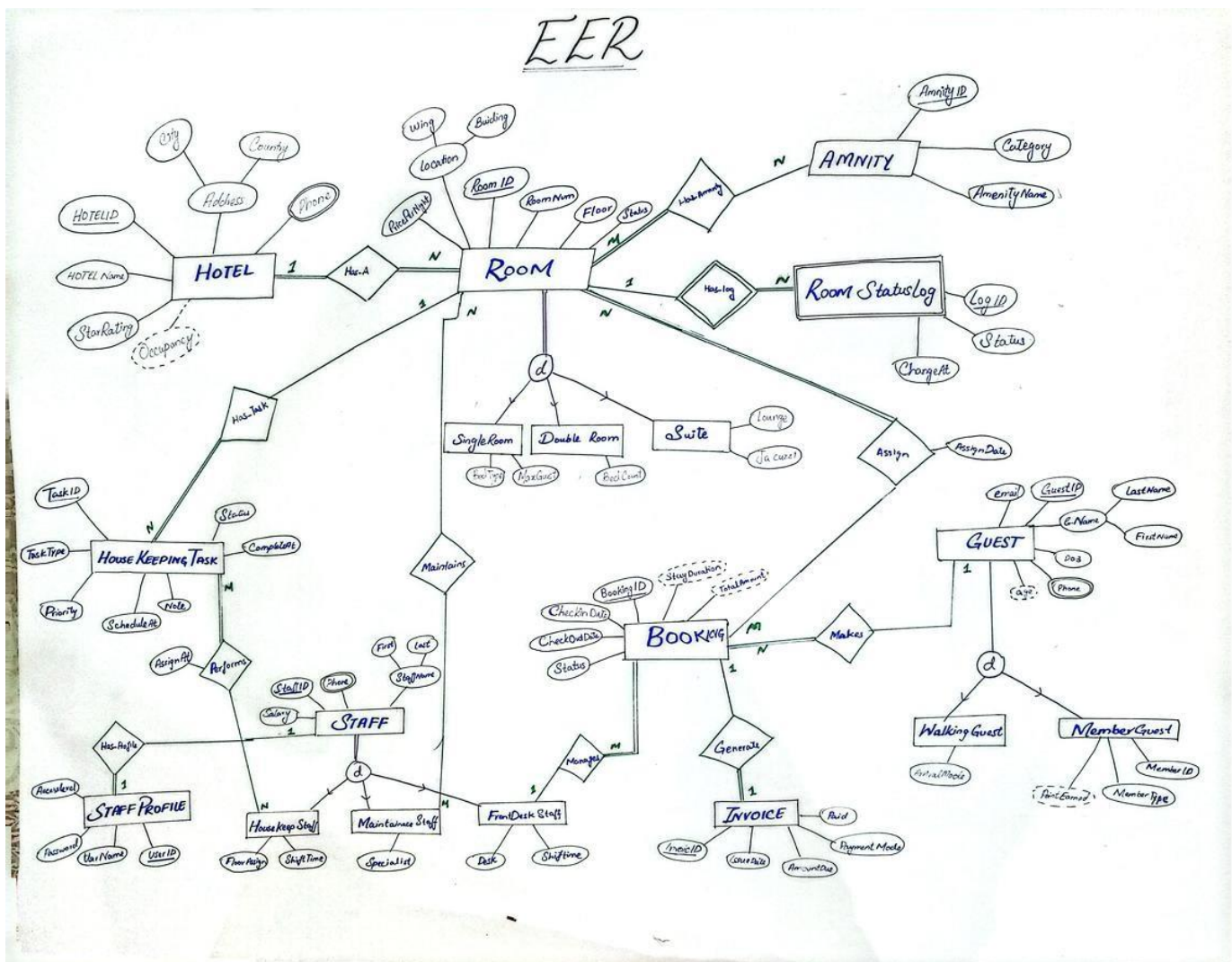
- This system also mirrors the way hotels do business in the real world; Hotels are categorized into different classes of quality with different pricing and types of rooms.
 - o Guests can be regular or member guests.
 - o Staff roles are specialized.
 - o Bookings link guests to their hotels.
 - o Payments are made using an invoice.
- o Usage of the hotel room is tracked in a log.
- o Housekeeping has to maintain the condition of the hotel room, etc.

3. Conclusion

The EER model integrates Entity Relationship (ER) concepts (Entities, attributes, relationships) with Enhanced Entity Relationship (EER) concepts: specialization, weak entities, multivalued and composite attributes. The design achieves real-world accuracy while providing acceptable implementation efficiency for relational (Oracle) and NoSQL (MongoDB) databases. The design avoids redundancy, maintains data integrity, and supports efficient querying and analysis.

PART-I

ENANCED ENTITY-RELATIONSHIP DIAGRAM



RELATIONAL SCHEMA

Hotel (HotelID (Primary key), HotelName, City, Country, StarRating)

⇒ Occupancy is a derived attribute.

Room (RoomID (Primary key), RoomNum, Floor, Status, Wing, Building, PricePerNight, HotelID (Foreign key))

Amenity (AmenityID (Primary key), category, AmenityName)

Has-Amenity (RoomID, AmenityID)

Composite key = RoomID + AmenityID

RoomStatusLog (logID (Partial key), RoomID, status, changeAt)

Composite key = RoomID + logID

Hotel-Phone (HotelID, Phone)

Composite key = HotelID + Phone

Maintains-Room (RoomID, staffID)

Composite key = RoomID + staffID

SingleRoom (RoomID (Primary key), BedType, MaxGuest)

DoubleRoom (RoomID (Primary key), BedCount)

Suite (RoomID (Primary key), Lounge, Jacuzzi)

DATE: ___/___/___

Guest (GuestID (Primary key), email, DOB, First Name, LastName)

→ Age is a derived attribute

Guest-Phone (GuestID, phone)

Composite key: GuestID + Phone

MemberGuest (GuestID (Primary key), MemberID, MemberType)

→ PointsEarned is a derived attribute

WalkingGuest (GuestID (Primary key), Arrival Mode)

Booking (BookingID (Primary key), CheckInDate, CheckOutDate, Status, StaffID (foreign key), GuestID (foreign key))

Assign-Room (BookingID, RoomID, AssignDate)

Composite key: BookingID + RoomID

Invoice (InvoiceID (Primary key), IssueDate, AmountDue, PaymentMade, Paid, BookingID (Foreign key))

Staff (StaffID (Primary key), FirstName, LastName, Salary)

Staff-Phone (StaffID, phone)

Composite key: StaffID + Phone

DATE: ___/___/___

Staff-Profile (UserID (Primary Key), UserNames,
Password, AccessLevel, StaffID (Foreign Key))

Housekeep Staff (StaffID (Primary Key), FloorAssign,
ShiftTime)

Maintenance Staff (StaffID (Primary key),
Specialist)

FrontDesk Staff (StaffID (Primary key), Desk,
ShiftTime)

House Keeping Task (TaskID (Primary Key), TaskType,
Priority, Schedule At, Note, Status, Complete At,
RoomID (foreign key))

Performs-Task (TaskID, StaffID, AssignAt)
Composite key: StaffID + TaskID

PART 1a:

Users > Kashan > Downloads > db_project > PROJECT_SQL (1).sql

```
1 CREATE TABLE HOTEL (  
2 HotelID INT PRIMARY KEY,  
3 HotelName VARCHAR2(100) NOT NULL,  
4 City VARCHAR2(50),  
5 Country VARCHAR2(50),  
6 StarRating NUMBER(1) CHECK (StarRating BETWEEN 1 and 5)  
7 );  
8 INSERT INTO Hotel VALUES (1, 'Grand Palace Hotel', 'Islamabad', 'Pakistan', 5);  
9 INSERT INTO Hotel VALUES (2, 'City Comfort Inn', 'Lahore', 'Pakistan', 3);  
10 INSERT INTO Hotel VALUES (3, 'Royal Stay', 'Karachi', 'Pakistan', 4);  
11 INSERT INTO Hotel VALUES (4, 'Blue Horizon', 'Balochistan', 'Pakistan', 5);  
12 INSERT INTO Hotel VALUES (5, 'Mountain View Lodge', 'Murree', 'Pakistan', 3);  
13  
14 CREATE TABLE Hotel_Phone (  
15 HotelID INT,  
16 Phone VARCHAR2(20),  
17 PRIMARY KEY (HotelID, Phone),  
18 FOREIGN KEY (HotelID) REFERENCES Hotel(HotelID)  
19 );  
20 INSERT INTO Hotel_Phone VALUES (1, '+92-51-1234567');  
21 INSERT INTO Hotel_Phone VALUES (1, '+92-51-7654321');  
22 INSERT INTO Hotel_Phone VALUES (2, '+92-42-9876543');  
23 INSERT INTO Hotel_Phone VALUES (2, '+92-42-1112233');  
24 INSERT INTO Hotel_Phone VALUES (3, '+92-21-5556677');  
25 INSERT INTO Hotel_Phone VALUES (4, '+92-43-5556677');  
26 INSERT INTO Hotel_Phone VALUES (4, '+92-41-9998877');  
27 INSERT INTO Hotel_Phone VALUES (5, '+92-51-3334455');  
28 INSERT INTO Hotel_Phone VALUES (5, '+92-51-6667788');
```

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Hasnback | Dependencies | Details | Partitions | Indexes | SQL

   Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	HOTELID	NUMBER(38,0)	No	(null)	1	(null)
2	HOTELNAME	VARCHAR2(100 BYTE)	No	(null)	2	(null)
3	CITY	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	COUNTRY	VARCHAR2(50 BYTE)	Yes	(null)	4	(null)
5	STARRATING	NUMBER(1,0)	Yes	(null)	5	(null)

```

INSERT INTO WALKING_GUEST VALUES (8, 'Walk-in');
CREATE TABLE STAFF (
STAFFID INT PRIMARY KEY ,
S_FIRST_NAME VARCHAR2 (40) NOT NULL ,
S_LAST_NAME VARCHAR2 (40),
S_SALARY INT CHECK (S_SALARY>2000)
);
INSERT INTO STAFF VALUES (1, 'Usman', 'Khan', 55000);
INSERT INTO STAFF VALUES (2, 'Ayesha', 'Malik', 48000);
INSERT INTO STAFF VALUES (3, 'Bilal', 'Ahmed', 62000);
INSERT INTO STAFF VALUES (4, 'Sana', 'Raza', 45000);
INSERT INTO STAFF VALUES (5, 'Kamran', 'Sheikh', 70000);
INSERT INTO STAFF VALUES (6, 'Nadia', 'Hassan', 53000);
INSERT INTO STAFF VALUES (7, 'Tariq', 'Butt', 41000);
INSERT INTO STAFF VALUES (8, 'Farrukh', 'Qureshi', 38000);
INSERT INTO STAFF VALUES (9, 'Zaid', 'Mirza', 44000);
INSERT INTO STAFF VALUES (10, 'Hasan', 'Naqvi', 46000);
INSERT INTO STAFF VALUES (11, 'Omair', 'Siddiqui', 50000);
INSERT INTO STAFF VALUES (12, 'Rehan', 'Javed', 43000);
INSERT INTO STAFF VALUES (13, 'Asad', 'Gillani', 47000);
INSERT INTO STAFF VALUES (14, 'Faisal', 'Chaudhry', 49000);
INSERT INTO STAFF VALUES (15, 'Junaid', 'Ansari', 42000);
INSERT INTO STAFF VALUES (16, 'Waqar', 'Mehmood', 45000);
INSERT INTO STAFF VALUES (17, 'Hira', 'Baig', 40000);
INSERT INTO STAFF VALUES (18, 'Maham', 'Iqbal', 41000);
INSERT INTO STAFF VALUES (19, 'Sobia', 'Rehman', 39000);
INSERT INTO STAFF VALUES (20, 'Anum', 'Shah', 40000);
INSERT INTO STAFF VALUES (21, 'Rimsha', 'Aslam', 38000);
INSERT INTO STAFF VALUES (22, 'Daniyal', 'Farooq', 42000);
INSERT INTO STAFF VALUES (23, 'Saad', 'Zafar', 39000);
INSERT INTO STAFF VALUES (24, 'Mehak', 'Yousaf', 40000);

```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	STAFFID	NUMBER (38, 0)	No	(null)	1 (null)	
2	S_FIRST_NAME	VARCHAR2 (40 BYTE)	No	(null)	2 (null)	
3	S_LAST_NAME	VARCHAR2 (40 BYTE)	Yes	(null)	3 (null)	
4	S_SALARY	NUMBER (38, 0)	Yes	(null)	4 (null)	

PART 1B

Join Query

English Requirement

Show guest name with their booking details.

Worksheet

Query Builder

```

SELECT G.GUESTID,
       G.FIRST_NAME,
       G.LAST_NAME,
       B.BOOKINGID,
       B.STATUS,
       B.CHECK_DATE,
       B.CHECKOUT_DATE
FROM GUEST G
JOIN BOOKING B
ON G.GUESTID = B.GUESTID;

```

Query Result x

All Rows Fetched: 8 in 0.214 seconds

	GUESTID	FIRST_NAME	LAST_NAME	BOOKINGID	STATUS	CHECK_DATE	CHECKOUT_DATE
1	1	Ali	Hassan	1	Confirmed	10-JAN-26	15-JAN-26
2	2	Sara	Khan	2	Confirmed	01-FEB-26	05-FEB-26
3	3	John	Doe	3	Checked-Out	10-FEB-26	14-FEB-26
4	4	Amna	Baig	4	Confirmed	01-MAR-26	07-MAR-26
5	5	Raza	Qureshi	5	Cancelled	15-MAR-26	20-MAR-26
6	6	Maria	Ali	6	Confirmed	01-APR-26	03-APR-26
7	7	Hamza	Tariq	7	Confirmed	10-APR-26	15-APR-26
8	8	Zara	Anwar	8	Confirmed	01-MAY-26	04-MAY-26

Subquery

English Requirement

Find hotels whose average room price is higher than the overall average room price.

Worksheet | Query Builder

```
SELECT HotelID, HotelName
FROM HOTEL
WHERE HotelID IN (
    SELECT HotelID
    FROM ROOM
    GROUP BY HotelID
    HAVING AVG(PricePerNight) >
        (SELECT AVG(PricePerNight) FROM ROOM)
);
```

Query Result x

SQL | All Rows Fetched: 2 in 0.036 seconds

	HOTELID	HOTELNAME
1	1	Grand Palace Hotel
2	4	Blue Horizon

Set Operator

English Requirement

Find guests who are in both MEMBER_GUEST and WALKING_GUEST tables.

Worksheet

Query Builder

SELECT GUESTID FROM MEMBER_GUEST

UNION

SELECT GUESTID FROM WALKING_GUEST;

Query Result x

SQL | All Rows Fetched: 8 in 0.007 seconds

	GUESTID
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8

PART-II

```
staff_profile.json x maintainsroom.json
> Users > Kashan > Downloads > db_project > {} staff_profile.json
1  [
2  { "userid": 1, "username": "ahmed.khan", "password": "$2b$12$KIXab", "accesslevel": "Admin", "staffid": 1 },
3  { "userid": 2, "username": "sara.malik", "password": "$2b$12$LJY", "accesslevel": "Manager", "staffid": 2 },
4  { "userid": 3, "username": "usman.ali", "password": "$2b$12$MK", "accesslevel": "FrontDesk", "staffid": 3 },
5  { "userid": 4, "username": "hina.raza", "password": "$2b$12$NLA", "accesslevel": "FrontDesk", "staffid": 4 },
6  { "userid": 5, "username": "bilal.queshi", "password": "$2b$12$OMB", "accesslevel": "Housekeeping", "staffid": 5 },
7  { "userid": 6, "username": "zara.hussain", "password": "$2b$12$PN", "accesslevel": "Housekeeping", "staffid": 6 },
8  { "userid": 7, "username": "tariq.mehmood", "password": "$2b$12$QOD", "accesslevel": "Maintenance", "staffid": 7 },
9  { "userid": 8, "username": "nadia.shah", "password": "$2b$12$RPE", "accesslevel": "Manager", "staffid": 8 }
10 ]
11
```

Edit Explain Run Options

Showing 1 – 20 count results

Create Collection

Collection Name

☐ Time-Series
Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences (e.g. Custom collation, Clustered collections)

Cancel Create Collection

```
staff_profile.json amenity (1).json X maintainsroom.json
C: > Users > Kashan > Downloads > db_project > amenity (1).json > ...
1
2 { "amenityid": 1, "category": "Entertainment", "amenityname": "Flat Screen TV" },
3 { "amenityid": 2, "category": "Comfort", "amenityname": "Air Conditioning" },
4 { "amenityid": 3, "category": "Comfort", "amenityname": "King Size Bed" },
5 { "amenityid": 4, "category": "Hygiene", "amenityname": "Private Bathroom" },
6 { "amenityid": 5, "category": "Luxury", "amenityname": "Jacuzzi" },
7 { "amenityid": 6, "category": "Connectivity", "amenityname": "Free WiFi" },
8 { "amenityid": 7, "category": "Food", "amenityname": "Mini Bar" },
9 { "amenityid": 8, "category": "Luxury", "amenityname": "Lounge Area" },
10 { "amenityid": 9, "category": "Comfort", "amenityname": "Balcony" },
11 { "amenityid": 10, "category": "Hygiene", "amenityname": "Bathtub" }
12
13
```

lookup Edit

EXPORT DATA

Create Collection

Collection Name

☐ Time-Series
Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences (e.g. Custom collation, Clustered collections)

Cancel Create Collection

: ObjectId('6a...
ffID : 1
stName : "Usma...
tName : "Khan"
ary : 55000
nes : Array (1
el_info : Arra...
: ObjectId('6a...
ffID : 2
stName : "Ayes...
tName : "Malik...
ary : 48000
nes : Array (1
el_info : Arra...
: ObjectId('6a09b9d694595cb630903944')
ffID : 3

PART-III

MONGO DB DATABASE :

MongoDB Compass - localhost:27017/hotel_management

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

hotel_management

amenity

booking

double_room

frontdesk_staff

guest

guest_phone

has_amnety

hotel

hotel_phone

house_keeping_staff

housekeeping_task

invoice

maintainsroom

maintenance_Staff

member_guest

perform_task

localhost:27017 > hotel_management

Open MongoDB shell

Create collection

Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
amenity	-	20.48 kB	884.00 B	10	88.00 B	1	20.48 kB
booking	-	20.48 kB	1.62 kB	8	202.00 B	1	20.48 kB
double_room	-	20.48 kB	192.00 B	4	48.00 B	1	20.48 kB
frontdesk_staff	-	20.48 kB	711.00 B	8	88.00 B	1	20.48 kB
guest	-	36.86 kB	1.42 kB	11	129.00 B	1	36.86 kB
guest_phone	-	20.48 kB	544.00 B	8	68.00 B	1	20.48 kB
has_amnety	-	20.48 kB	833.00 B	17	49.00 B	1	20.48 kB
hotel	-	20.48 kB	614.00 B	5	122.00 B	1	20.48 kB
hotel_phone	-	20.48 kB	549.00 B	9	61.00 B	1	20.48 kB
house_keeping_staff	-	20.48 kB	695.00 B	8	86.00 B	1	20.48 kB
housekeeping_task	-	20.48 kB	1.65 kB	8	206.00 B	1	20.48 kB
invoice	-	20.48 kB	1.06 kB	8	132.00 B	1	20.48 kB
maintainsroom	-	20.48 kB	376.00 B	8	47.00 B	1	20.48 kB
maintenance_Staff	-	20.48 kB	487.00 B	8	60.00 B	1	20.48 kB
member_guest	-	20.48 kB	363.00 B	5	72.00 B	1	20.48 kB
perform_task	-	20.48 kB	376.00 B	8	47.00 B	1	20.48 kB

TABLE BOOKING

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

hotel_management

amenity

booking

double_room

frontdesk_staff

guest

guest_phone

has_amnety

hotel

hotel_phone

house_keeping_staff

housekeeping_task

invoice

maintainsroom

maintenance_Staff

member_guest

perform_task

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25

1 - 8 of 8

```
{
  "_id": ObjectId('6a084695f182c2e2e4abb769'),
  "bookingid": 1,
  "check_date": "2026-01-10",
  "checkout_date": "2026-01-15",
  "status": "Confirmed",
  "staffid": 2,
  "guestid": 1,
  "assigned_room": Object
}
```

```
{
  "_id": ObjectId('6a084695f182c2e2e4abb76a'),
  "bookingid": 2,
  "check_date": "2026-02-01",
  "checkout_date": "2026-02-05",
  "status": "Confirmed",
  "staffid": 2,
  "guestid": 2,
  "assigned_room": Object
}
```

```
{
  "_id": ObjectId('6a084695f182c2e2e4abb76b'),
  "bookingid": 3,
  "check_date": "2026-02-10",
  "checkout_date": "2026-02-14",
  "status": "Checked-Out",
  "staffid": 4,
  "guestid": 3,
  "assigned_room": Object
}
```

Page 11 of 11 1188 words Text Predictions: On Accessibility: Unavailable

Focus

26°C Partly sunny

8:21 AM

5/3/2025

TABLE _ GUEST :

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- booking
- double_room
- frontdesk_staff
- guest
- guest_phone
- has_amnety
- hotel
- hotel_phone
- house_keeping_staff
- housekeeping_task
- invoice
- maintainsroom
- maintenance_Staff
- member_guest
- perform_task
- room
- roomstatuslog
- single_room
- staff
- staff_phone
- staff_profile

localhost:27017 > hotel_management > guest

Documents 11 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE

```
_id: ObjectId('6a084926f182c2e2e4abb775')
guestid: 1
first_name: "Ali"
last_name: "Hassan"
email: "ali.hassan@gmail.com"
dob: "1990-05-14"
```

```
_id: ObjectId('6a084926f182c2e2e4abb776')
guestid: 2
first_name: "Sara"
last_name: "Khan"
email: "sara.khan@hotmail.com"
dob: "1985-11-22"
```

```
_id: ObjectId('6a084926f182c2e2e4abb777')
guestid: 3
first_name: "John"
last_name: "Doe"
email: "john.doe@gmail.com"
dob: "1992-03-08"
```

```
_id: ObjectId('6a084926f182c2e2e4abb778')
guestid: 4
first_name: "Amna"
last_name: "Baig"
email: "amna.baig@yahoo.com"
dob: "1998-07-30"
```

Cursor is AI and can make mistakes. Please double-check responses.

PART-IV

GUI :

Hotel Management System

• Connected

Guest Collection
Manage all registered guests

	GUESTID	FIRST NAME	LAST NAME	EMAIL	DOB
1		Ali	Hassan	ali.hassan@gmail.com	1990-05-14
2		Sara	Khan	sara.khan@hotmail.com	1985-11-22
3		John	Doe	john.doe@gmail.com	1992-03-08
4		Amna	Baig	amna.baig@yahoo.com	1998-07-30
5		Raza	Qureshi	raza.qureshi@gmail.com	1978-01-17
6		Maria	Ali	maria.ali@outlook.com	2000-09-05
7		Hamza	Tariq	hamza.tariq@gmail.com	1995-12-25
8		Zara	Anwar	zara.anwar@gmail.com	1988-04-11
9		imara	asim	imaraasim01@gmail.com	2006-11-14
10		umaima	baloch	umaima0@gmail.com	2005-04-04
11		shafia	shafique	shafial@gmail.com	2005-12-18

Refresh Insert Update Delete

CRUD OPERATIONS

insertion :

The screenshot shows a web application titled "Hotel Management System". A modal window titled "Insert Guest" is open, displaying a form with the following fields and values:

- guestid**: 12
- first_name**: laiba
- last_name**: wajid
- email**: laiba@gmail.com
- dob**: 2006-01-27

A green "Submit" button is located at the bottom of the form. In the background, a table is partially visible with columns for "Amenity" and "Member Guest". The table contains the following data:

Amenity	Member Guest
9	imara
10	umaima
11	shafia

data insertion (in MongoDB Campus):

```
email : "umaima@gmail.com"  
dob : "2005-04-04"
```

```
▶ _id: ObjectId('6a09bb6f0681cbf42b5ca201')  
guestid : 11  
first_name : "shafia"  
last_name : "shafique"  
email : "shafia@gmail.com"  
dob : "2005-12-18"
```

```
_id: ObjectId('6a0e9f82b90481368fda0915')  
guestid : 12  
first_name : "laiba"  
last_name : "wajid "  
email : "laiba@gmail.com"  
dob : "2006-01-27"
```

tions: On Accessibility: Investigate

Focus

update :

Update Guest

first_name

laiba

last_name

khan

email

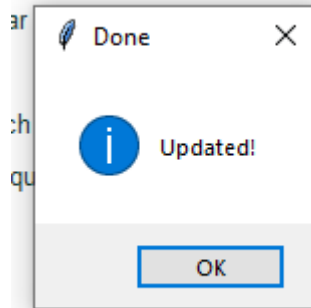
laiba@gmail.com

dob

2006-01-27

Submit

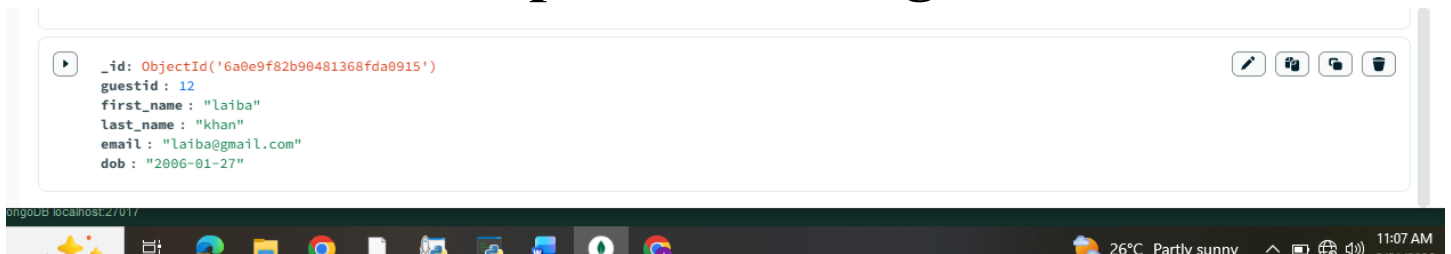
9 imara



data update in gui :

11	Shadia	Shadique	Shadia@gmail.com	2006-12-10
12	laiba	khan	laiba@gmail.com	2006-01-27

data update in mongoDB :



Delete :

3

4

5

6

7

8

9

10

11

12

John

Amna

Raza

Maria

Hamza

Zara

imara

umaima

shafia

laiba

Doe

Baig

Qureshi

Ali

Tariq

Al

a

b

s

k

john.doe@gmail.com

amna.baig@yahoo.com

raza.qureshi@gmail.com

maria.ali@outlook.com

hamza.tariq@gmail.com

zara.anwar@gmail.com

imaraasim01@gmail.com

umaima0@gmail.com

shafial@gmail.com

laiba@gmail.com

1992-03-08

1998-07-30

1978-01-17

2000-09-05

1995-12-25

1988-04-11

2006-11-14

2005-04-04

2005-12-18

2006-01-27

Confirm

?

Delete guest ID 12?

Yes

No

Refresh

Insert

Update

Delete

Data deleted from gui :

9

10

11

imara

umaima

shafia

asim

baloch

shafique

imaraasim01@gmail.com

umaima0@gmail.com

shafial@gmail.com

2006-11-14

2005-04-04

2005-12-18

Data deleted from mongoDB :

The screenshot shows the MongoDB Compass interface. At the top, there is a query bar with the text "Type a query: { field: 'value' } or [Generate query](#)". To the right of the query bar are buttons for "Explain", "Reset", "Find", and "Options". Below the query bar is a row of action buttons: "ADD DATA", "UPDATE", "DELETE", "EXPORT DATA", and "EXPORT CODE". To the right of these buttons is a status bar showing "25" and "1 - 11 of 11".

The main area displays a list of documents. The first document is:

```
{
  "_id": ObjectId("600720120c6e2e90071c"),
  "guestid": 8,
  "first_name": "Zara",
  "last_name": "Anwar",
  "email": "zara.anwar@gmail.com",
  "dob": "1988-04-11"
}
```

The second document is highlighted:

```
{
  "_id": ObjectId('6a0894e19bdeca079762079f'),
  "guestid": 9,
  "first_name": "imara",
  "last_name": "asim",
  "email": "imaraasim01@gmail.com",
  "dob": "2006-11-14"
}
```

The third document is:

```
{
  "_id": ObjectId('6a08964a11f4093298995f19'),
  "guestid": 10,
  "first_name": "umaima",
  "last_name": "baloch",
  "email": "umaima0@gmail.com",
  "dob": "2005-04-04"
}
```

The fourth document is:

```
{
  "_id": ObjectId('6a09bb6f0681cbf42b5ca201'),
  "guestid": 11,
  "first_name": "shafia",
  "last_name": "shafique",
  "email": "shafia!@gmail.com",
  "dob": "2005-12-18"
}
```

The bottom of the screenshot shows a taskbar with various application icons and a system clock displaying "11:09 AM".

PART-V

AGGREGATE FUNCTION :

Documents 8AggregationsSchemaIndexes 1Validation

\$group

Generate aggregation + ? ExplainRunOptions ▶

Untitled – modifiedSAVE + CREATE NEWEXPORT DATAEXPORT CODEPREVIEWSTAGESTEXTWIZARD

8 Documents in the collection

Preview of documents

_id: ObjectId('6a09b9bc94595cb630903938')

InvoiceID : 1

IssueDate : "2024-01-05"

AmountDue : 450

PaymentMode : "Credit Card"

Paid : 1296

BookingID : 1

_id: ObjectId('6a09b9bc94595cb630903939')

InvoiceID : 2

IssueDate : "2024-01-12"

AmountDue : 820.5

PaymentMode : "Cash"

Paid : 8296

BookingID : 2

_id: ObjectId('6a09b9bc94595cb63090393a')

InvoiceID : 3

IssueDate : "2024-02-03"

AmountDue : 1200

PaymentMode : "Bank Transfer"

Paid : 3456

BookingID : 3

Stage 1 \$group

1 {
2 _id: null,
3 total: {
4 \$sum: "\$AmountDue"
5 }
6 }

Output preview after \$group stage (Sample of 1 document)

_id: null

total : 7157

RESULT

Documents 8AggregationsSchemaIndexes 1Validation

\$group

Edit

Collation { locale: 'simple' }

ALL RESULTSEXPORT DATAEXPORT CODE

_id: null

total : 7157

QUERY2 :

Stage1

\$group

1

{

2

"_id": "\$PaymentMode",

3

"total": {

4

"\$sum": "\$AmountDue"

5

},

6

"count": {

7

"\$sum": 1

8

}

9

}

Output preview after \$group stage (Sample of 4 documents)

_id: "Credit Card"

total : 1100

count : 2

_id: "Cash"

total : 1800.75

count : 2

\$group

Edit

ALL RESULTS

EXPORT DATA

EXPORT CODE

_id: "Bank Transfer"

total : 3350

count : 2

_id: "Debit Card"

total : 906.25

count : 2

_id: "Credit Card"

total : 1100

count : 2

_id: "Cash"

total : 1800.75

count : 2

GUI

:

insertion

:

data

insertion

:

```
email : "umaima@gmail.com"
dob : "2005-04-04"
```

```
_id: ObjectId('6a09bb6f0681cbf42b5ca201')
guestid : 11
first_name : "shafia"
last_name : "shafique"
email : "shafia@gmail.com"
dob : "2005-12-18"
```

```
_id: ObjectId('6a0e9f82b90481368fda0915')
guestid : 12
first_name : "laiba"
last_name : "wajid "
email : "laiba@gmail.com"
dob : "2006-01-27"
```

tions: On Accessibility: Investigate

Focus

JOIN

Write a MongoDB aggregation query to join the rooms collection with the hotels collection using the common field hotelid, and store the matched hotel details inside a field named hotel_info.

Stage1 \$lookup

```
1
2 {
3   "from": "hotels",
4   "localField": "hotelid",
5   "foreignField": "hotelid",
6   "as": "hotel_info"
7 }
8
9
```

Output preview after \$lookup stage (Sample of 10 documents)

```
_id: ObjectId('6a09b9d694595cb630903942')
StaffID : 1
FirstName : "Usman"
LastName : "Khan"
Salary : 55000
Phones : Array (1)
hotel_info : Array (empty)
```

```
_id: ObjectId('6a09b9d694595cb630903942')
StaffID : 2
FirstName : "Ayesha"
LastName : "Malik"
Salary : 48000
Phones : Array (1)
hotel_info : Array (empty)
```

RESULTS

:

```
▶ _id: ObjectId('6a09b9d694595cb630903942')
StaffID : 1
FirstName : "Usman"
LastName : "Khan"
Salary : 55000
▶ Phones : Array (1)
▶ hotel_info : Array (empty)
```

```
_id: ObjectId('6a09b9d694595cb630903943')
StaffID : 2
FirstName : "Ayesha"
LastName : "Malik"
Salary : 48000
▶ Phones : Array (1)
▶ hotel_info : Array (empty)
```

```
_id: ObjectId('6a09b9d694595cb630903944')
StaffID : 3
FirstName : "Bilal"
LastName : "Ahmed"
Salary : 62000
▶ Phones : Array (1)
▶ hotel_info : Array (empty)
```

```
_id: ObjectId('6a09b9d694595cb630903945')
StaffID : 4
```

